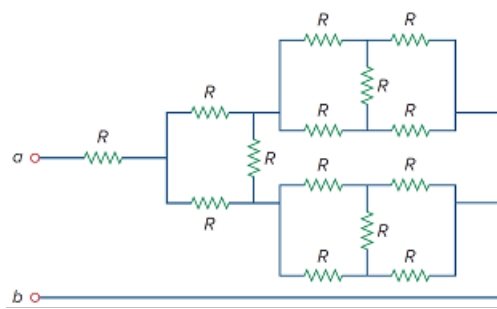


Problem

Find R_{ab} in the four-way power divider circuit in Fig. 2.135. Assume each $R = 4\ \Omega$.

Figure 2.135:



Step-by-step solution

Step 1 of 9

Refer to Figure 2.135 in the textbook.

Convert the delta sub networks into wye networks. Draw the modified circuit diagram.

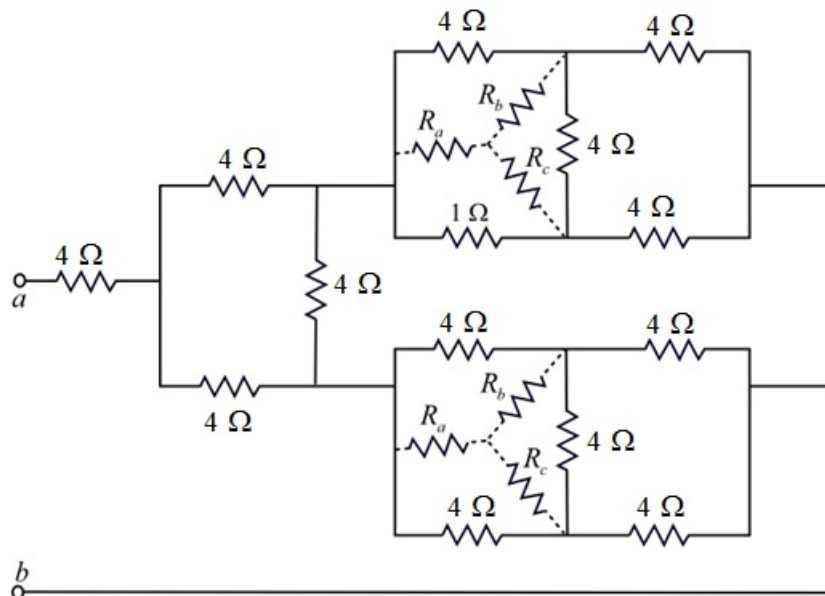


Figure 1

Step 2 of 9

Calculate the star connected resistances.

$$R_a = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

$$R_b = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

$$R_c = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

Step 3 of 9

Draw the modified circuit diagram.

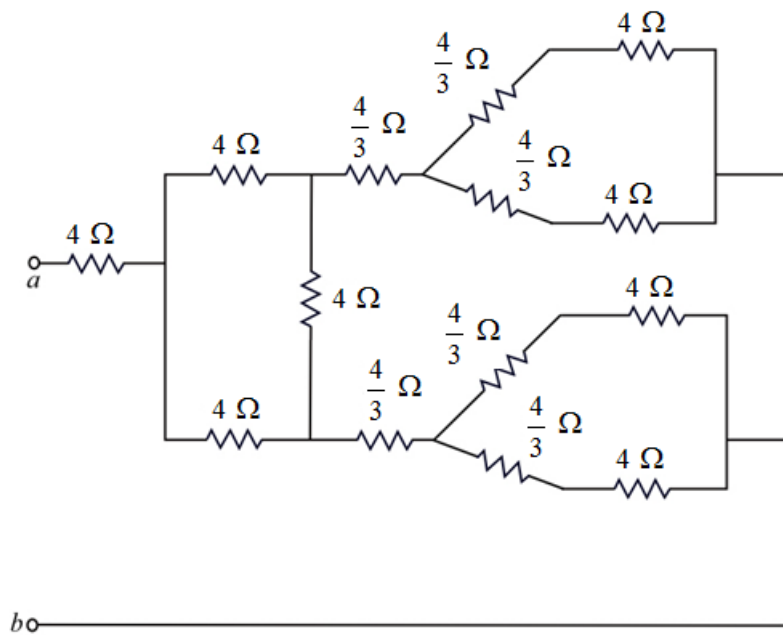


Figure 2

Step 4 of 9

From the Figure 2, calculate the equivalent resistance of right side top sub network.

$$R_{eq} = \left(\frac{4}{3} + 4 \right) \parallel \left(\frac{4}{3} + 4 \right) + \frac{4}{3}$$

$$= \left(\frac{16}{3} \right) \parallel \left(\frac{16}{3} \right) + \frac{4}{3}$$

$$= \frac{\left(\frac{16}{3} \right) \left(\frac{16}{3} \right)}{\frac{16}{3} + \frac{16}{3}} + \frac{4}{3}$$

$$= \frac{8}{3} + \frac{4}{3}$$

$$= 4 \Omega$$

Similarly, the bottom right side network can be replaced with 4Ω resistor.

Step 5 of 9

Draw the modified circuit diagram.

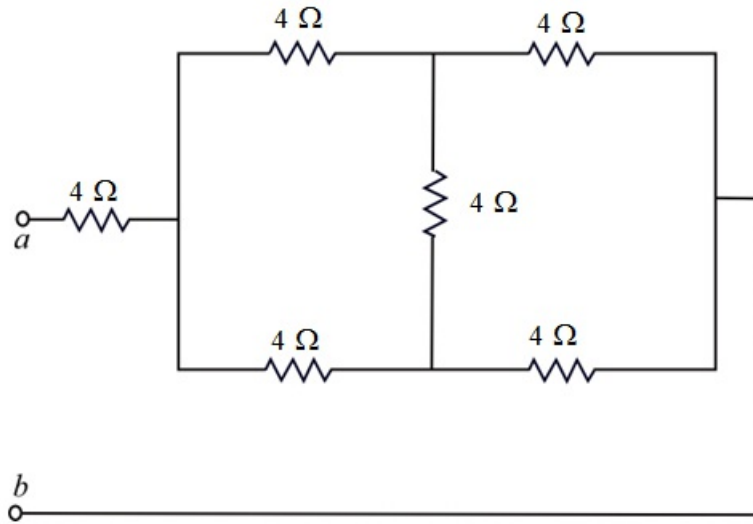


Figure 3

Step 6 of 9

Convert the delta sub network into wye network as shown in the following figure.

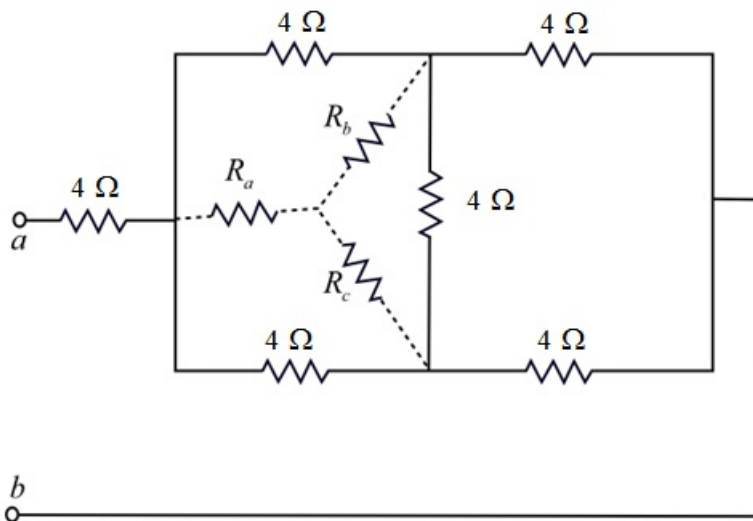


Figure 4

Step 7 of 9

Calculate the star connected resistances.

$$R_a = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

$$R_b = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

$$R_c = \frac{(4)(4)}{4+4+4}$$

$$= \frac{4}{3} \Omega$$

Step 8 of 9

Draw the modified circuit diagram.

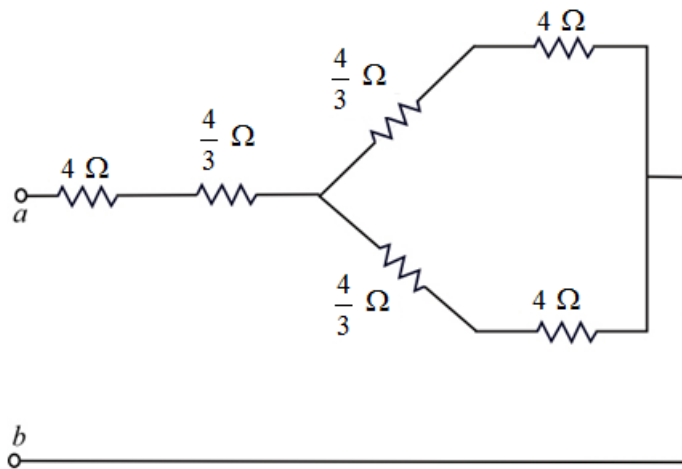


Figure 5

Step 9 of 9

Calculate the equivalent resistance across ab terminals.

$$R_{ab} = \left(\frac{4}{3} + 4 \right) \parallel \left(\frac{4}{3} + 4 \right) + \frac{4}{3} + 4$$

$$= \left(\frac{16}{3} \right) \parallel \left(\frac{16}{3} \right) + \frac{4}{3} + 4$$

$$= \frac{8}{3} + \frac{4}{3} + 4$$

$$= 4 + 4$$

$$= 8 \Omega$$

Therefore, the equivalent resistance across ab terminals, R_{ab} is $\boxed{8 \Omega}$.